

Spatial Informatics (GS2.401)

Complete Course Study Guide

All Classes: 31 July – 19 September 2026

Compiled from slide decks, transcripts, handwritten notes & PYQs

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How to Use This Guide

This is the merged, cumulative guide covering **every class held so far** – all 12 sessions from **31 July** through **19 September 2026** – built by combining the Quiz 1 material (Introduction; GIS Data Models; Spatial Data Collection) with the Quiz 2 material (Data Cleaning & Topology; Vector Data Analysis; Raster Data Analysis). Content is grounded in the lecture slide decks (the primary exam source), cross-checked against the **Notes/** folder’s handwritten class notes, and supplemented by transcript-derived summaries where they add clarifying detail without contradicting the slides.

Section 8 mines **four** previous-year papers from the **Ref/** folder for recurring question patterns: two **Quiz 1** papers (a Monsoon-2024/2023-style paper, and – notably – **the actual Quiz 1 administered this year, 1 Sept 2026**) and two **Mid-Semester** papers (21 Sept 2024 and 25 Sept 2025) from the same instructor, Dr. P. Rama Chandra Prasad. Questions that require **map projections or the Geoid** are excluded throughout – projections/Geoid are not covered in any of the 12 classes to date – but everything else (including DGPS, BIP/BSQ/BIL, and raster/vector overlay reasoning) *is* in scope and is adapted into practice questions.

Study priority across the whole course so far

1. **Vector data structures:** spaghetti vs. vertex-dictionary (no topology) vs. DIME vs. arc-node (topology) – and what “topology” actually buys you.
2. **Vector overlay operations:** Clip vs. Intersect vs. Union vs. Erase/Difference – the attribute-handling distinction is the instructor’s favourite trap.
3. **Raster fundamentals:** grid structure, band interleaving (BSQ/BIL/BIP), compression (RLE/quadtrees), resolution trade-offs, and the Local/Focal/Zonal/Global operation classification.
4. **Data-collection error sources and fixes:** GPS/GNSS errors, digitizing errors (undershoot/overshoot/sliver/pseudonode), and the QGIS tools (snapping, topology checker) that fix them.
5. **The 5 topological relationship types** (connectivity, adjacency, containment, proximity, relative direction) and what survives a rubber-sheet stretch.
6. **Sensor platforms & resolutions:** satellite/aircraft/drone trade-offs; spatial/spectral/temporal/radiometric resolution; LiDAR first/last returns; SAR polarization.
7. **Point-in-polygon, overlay rules (Enumeration/Dominance/Contributory/Interaction), reclassification, and map generalization operators.**

1 31 July: Introduction to Spatial Informatics

1.1 Overview

The opening lecture frames the whole course around one tagline: **“Everything is Spatial.”** **Spatial Informatics** = spatial data *plus* informatics – not just collecting location data, but storing it, cleaning it, and running analysis/algorithms on it to answer a real question (routing, density, prediction). Whether data “counts” as spatial depends on the *systems view*: if location is not **explicitly encoded** (no coordinates, no polygon), the system cannot use it as spatial data, even if it semantically refers to a place.

1.2 Spatio-Temporal Data & Cognitive Maps

“Something always happens somewhere.” Location helps answer *what/when/where* (easy) and *why/how* (hard, valuable – cause-effect, nearby influencing factors). Spatial data is inherently **spatio-temporal**: every activity carries both location and time (a photo’s EXIF, a bus’s GPS log). Minkowski (1905): space and time only preserve independent reality *in union*. The “space-time aquarium” (Parkes & Thrift, 1980) draws a person’s daily trajectory as a path rising through a vertical time axis.

Classify the data – a favourite question type

Scenario	Classification
Daily electricity bills, 5000 households, one year	Temporal / non-spatial (no location)
GPS logs of city buses every 10 s	Spatio-temporal (explicit location + time)
Customer DB: name, age, income, spend	Non-spatial
Satellite scans a river basin monthly in monsoon	Spatio-temporal
Police records: locality + date/time of crime	Spatio-temporal
City land-use map, one-time 2018 survey	Spatial (static)

We each carry a **cognitive (mental) map** of places we know – when giving directions we abstract to relevant landmarks, not every tree, and mental maps vary person to person. This motivates getting spatial awareness into a digital, processable form.

1.3 Types of Geographic Questions GIS Answers

1. **Geographic location** – Where is it? Why here? How much is here?
2. **Geographic distribution** – Local or global? Clustered or dispersed? Where are the boundaries?
3. **Geographic association** – What else is nearby? What co-occurs? What is absent in its presence?

4. **Geographic interaction** – Is it linked to something else? How much interaction between locations?
5. **Geographic change** – Has it always been here? How has it changed over time/space?

1.4 History of Spatial Analysis & GIS

John Snow, 1854, Soho, London – the first spatial analysis

500 cholera deaths in 10 days. Prevailing theory: airborne transmission. Snow plotted deaths as a spatial bar chart on a street map; the dense cluster pointed to **one contaminated water pump** (Broad Street), served by a different water company than the surrounding area. Removing the pump handle helped end the outbreak. This predates the word “GIS” by over a century.

Roger Tomlinson – “father of GIS” – built the first operational GIS (*Canada Geographic Information System*, ~1960) for land-use planning and coined the term “geographic information system.” **Survey of India** has been the national mapping authority since the 18th century.

Five eras of geospatial technology evolution (a sequencing question is likely):

1. **Field survey & analog mapping** (pre-1990s) – total stations, paper maps, data shared physically.
2. **Sensing systems & digital image processing** (1990s) – aerial photogrammetry, remote sensing, change studies (forest degradation, sprawl).
3. **Digital maps – desktop to web** (2000s) – CAD/GIS, interactive maps, web mashups.
4. **Geo Services** (2010s) – location as a variable; standard web services (WMS, WFS, WCS, CSW); you send a region+time, the provider processes and returns output.
5. **GeoAI** (2020s+) – spatio-temporal data science, AI/ML, natural-language geospatial queries.

Arc: *Measurements & Maps* → *Digital Data* → *Web & Services* → *Users & Communities* → *Intelligence & Impact*.

1.5 Six Sources of Spatial Data

1. **Remote Sensing** (satellite/aerial) – synoptic view, repeated observation, large-area coverage.
2. **GNSS/GPS** – point locations, trajectories, high temporal resolution.
3. **Survey & Field Data** – high accuracy, but sparse and costly.
4. **Administrative/Government Data** – authoritative, structured, often aggregated (census, cadastral maps).
5. **Sensors & IoT Networks** – fixed locations + continuous time → spatio-temporal streams.
6. **Crowdsourced/Volunteered Geographic Information (VGI)** – OpenStreetMap, geo-tagged social posts; **the data consumer is also the data source**. *Pokémon GO* is presented as a case study: player GPS + movement + visited POIs + AR scans became one of the largest crowdsourced geospatial datasets.

1.6 Layered Architecture & Geometry

The real world is represented as an **overlay of thematic layers**; you never mix feature types in one layer (customers = points, streets = lines, parcels = polygons, stored separately). All representation rests on **geometry**: **Point** (0-D, an (x, y) pair – an ATM), **Line** (1-D, ordered (x, y) series – a river), **Polygon** (2-D, closed line, first point = last point – a district), and **Multi-polygon** (several polygons treated as one unit – India *including* its island territories).

Every map you produce needs: **Title**, **Legend**, **Scale (bar)**, **North arrow** (the most-forgotten element), plus source info and a coordinate grid. *Never just take a screenshot.*

1.7 Capability Levels & GIScience/Systems/Services

Four things you can do with spatial data: (1) Visualize/depict, (2) Query/locate/identify (interactive), (3) Analysis (relationships, cause-effect, heatmaps), (4) Spatial Modelling (predict/simulate).

GIScience	The science: query algorithms, patterns, data mining, modelling
GISystems	The tools/tech: storage & processing, formats, data structures
GIServices	The data/capability delivered to users, storage-agnostic

1.8 Practice & Review

Q1. [MC] A spreadsheet of restaurant names with no coordinate column is, from a systems perspective:

- (A) Spatial
- (B) Non-spatial
- (C) Spatio-temporal
- (D) Ratio-scale

Answer: (B) – location is not explicitly encoded, so the system cannot treat it as spatial even though restaurants semantically have locations.

Q2. [Short Answer] Why is John Snow’s 1854 map considered the first spatial analysis? *Answer:* He mapped cholera deaths geographically rather than relying on the airborne theory, revealing a spatial cluster around one water pump – demonstrating that plotting data *in space* can reveal a cause invisible in tabular data alone.

Q3. [Short Answer] Name the five eras of geospatial technology evolution in order. *Answer:* Field survey/analog mapping → sensing systems/digital image processing → digital maps (desktop to web) → Geo Services → GeoAI.

2 4, 7, 11 Aug: GIS Data Models

2.1 Overview

GIS is a database management system whose data is geographic information, supporting capture, edit, display, validate, manipulate, and store. **A GIS data model is a model of the earth:** it subdivides the earth's surface into characterizable entities. Every feature has **spatial data** (geometry/location) and **non-spatial/attribute data** (descriptive records), kept in separate files linked by a DBMS.

2.2 Levels of Abstraction: Reality → Conceptual → Logical → Physical

R-C-L-P, using a road as the running example

- **Reality** – the actual road: centreline, width, lanes, pavement, intersections.
- **Conceptual model** (human-oriented) – identify features *and their relationships*: Line (centreline, network), Area (shoulder), Attributes (speed limit, lanes); relationships like “distance between two,” “borders on.”
- **Logical model** – choose the representation (vector vs. raster); define tables/records and links, e.g. Road{Line-id, Area-id, Road No., Name} → Line{Line-id, S-Node, E-Node, Vertices, Length}.
- **Physical model** – actual on-disk storage: files, records, database structures.

2.3 Spatial Data Types by Phenomenon

- **Continuous**: elevation, rainfall, ocean salinity – amorphous, no natural boundary, needs raster.
- **Areas – Unbounded**: land use, market areas, soils (fuzzy transitions). **Bounded**: city/state boundaries, ownership parcels (crisp borders).
- **Networks**: roads, transmission lines, streams.
- **Points – Fixed**: wells, lamp posts. **Moving**: cars, wildlife.

2.4 Types of Geographical Data Models

Model	Application
CAD	Simple mapping/engineering drawing – no relationships between objects
Graphical (non-topological)	Simple cartographic maps, no relationships
Image	Image processing, simple grid analysis
Raster / Grid	Spatial analysis and modelling
Vector / Geo-relational / Topological	Operations on point/line/polygon features with relationships
Network	Connected-system analysis (roads, rivers)
TIN	Terrain analysis (3-D surface from triangles)
Object	All entity types with properties/behaviour; good for 3-D

2.5 Vector Data Model

Definition (frequently tested)

A vector is a quantity with a starting coordinate plus an associated displacement and direction. In the vector *data model*: **spatial locations are EXPLICIT** (stored coordinates); **relationships between entities are IMPLICIT** (must be derived/built as topology). Built from primitives – points, lines, areas – via analytical geometry.

Point (0-D): too small for line/area, stored as (x, y) +attributes; a **node** is a topological point where ≥ 2 arcs connect. **Line/Arc (1-D)**: too narrow for an area; stored by first/last node + attributes (e.g. rivers with FNODE#, TNODE#, LPOLY#, RPOLY#, LENGTH). **Polygon (2-D)**: closed sequence of lines (first point = last point) – distinct from an open **polyline**.

2.6 Attribute Measurement Scales

Very likely on any quiz

Scale	Type	Meaning	Valid ops	Example
Nominal	Qualitative	Names/categories, no order	$=, \neq$; count	Land-cover class, gender
Ordinal	Qualitative	Ranked, gaps not meaningful	$<, >$; ranking	Low/moderate/high biodiversity
Interval	Quantitative	Numeric, equal intervals, no true zero	$+, -$	Temperature in $^{\circ}\text{C}$, a year
Ratio	Quantitative	Numeric, true zero	$+, -, \times, \div$	Rainfall (mm), population, area
Cyclic	Special	Wraps around (direction-type)	circular stats	Wind direction, slope aspect

Gotcha: cyclic data (wind direction, slope aspect) doesn't fit any of the other four scales – it wraps ($360^\circ = 0^\circ$).

2.7 Vector Data Structures (7 Aug)

Non-topological vs. Topological – a core distinction

Structure	Topology?	Redundancy	Note
Spaghetti	No	High (shared edge stored twice)	Every feature is an independent coordinate list
Vertex dictionary	No	None	Two files: unique vertices; features as ordered vertex-ID lists. No coordinate duplication, but still no topology
DIME	Yes	None	US Census Bureau, 1968. Each segment stores a name, from/to node IDs, and left/right polygon IDs
Arc-Node	Yes	None	The dominant modern structure (“intelligent data structure”); Node table (ID, Easting, Northing), Arc table (ID, From-Node, To-Node, Left-Poly, Right-Poly), Polygon table (ID, Arc list)

Topology = the explicit storage of spatial relationships (adjacency, connectivity, containment) – needed for network routing, watershed delineation, overlay, and error-checking. **Planar enforcement:** polygons in a coverage must not overlap and must not leave gaps – every point in the plane belongs to exactly one polygon.

2.8 Data Symbolization

Bertin's graphical variables for varying point/line/area symbols: **size**, **value (lightness)**, **grain (texture)**, **colour**, **orientation**, **shape** – match the visual variable to the data type (e.g. size for quantities, colour hue for nominal categories).

2.9 Raster Data Model (11 Aug)

Definition

A **grid** defines geographic space as a matrix of identically-sized square cells; each cell holds a **single numeric value** for one attribute at that location. Defined by: rows, columns, cell size (resolution), origin, NODATA value. Typically 8 bits/cell → 256 possible values, though continuous data can be float.

Spatial relationships are **implicit** (derived from the regular grid – adjacency via 4- or 8-connectivity). A **classified raster** assigns each pixel to a class via a lookup table (1=forest, 2=cultivated, ...), losing the original continuous values but reducing size.

Visualization: single-band grayscale (elevation); **pseudo-colour** (a palette over one grayscale band, e.g. NDVI); **false-colour composite** (3 bands mapped to R/G/B – e.g. Sentinel-2 NIR-Red-Green → healthy vegetation appears bright red).

2.9.1 Band Interleaving – Storage Formats

Near-certain quiz item – appears in multiple PYQs

Format	Ordering	Best for
BSQ (Band Sequential)	Entire first band, then entire second band, ...	Processing one band at a time (a filter on just NIR)
BIL (Band Interleaved by Line)	For each row: all bands' values for that row, then next row	Line-by-line processing; a compromise
BIP (Band Interleaved by Pixel)	For each pixel: all bands together, then next pixel	Per-pixel multi-band math – e.g. $NDVI = (NIR - Red) / (NIR + Red)$; most I/O-efficient for this

2.9.2 Raster Compression

- **Cell-by-cell array** – no compression; typical for float continuous data (DEM), where neighbours rarely match exactly.
- **Run-Length Encoding (RLE)** – encode runs as (count, value) pairs along each row; lossless; excellent when many neighbouring cells share a value (classified/low-res data), useless for a DEM.
- **Quadtree** – recursively subdivide into NW/NE/SW/SE quadrants, stopping where a block is homogeneous; compact for large uniform regions.

Worked RLE example

Row: C C C C C B B D D (9 cells) → RLE pairs: (5,C) (2,B) (2,D) – 9 raw values become 3 pairs (6 numbers). Applied row-by-row across a whole grid, RLE shrinks a classified raster dramatically when runs are long, but a checkerboard-like image with no runs would *not* compress at all.

2.9.3 Resolution vs. Detail vs. Storage

Smaller pixel (finer resolution) → more spatial detail, but many more pixels → larger file, more processing. Larger pixel (coarser) → less detail but compact and fast. **“Raster needs less processing than vector but consumes more storage.”** Raster does *not* provide precise locational information (DeMers, 1997) – undesirable for cadastral/legal work.

2.10 Vector vs. Raster – Comparison & Conversion

“Raster is faster, but vector is corrector.” – Joseph Berry

Aspect	Vector	Raster
Location	Explicit coordinates, precise at all scales	Implicit (row/col); precision limited by cell size
Relationships	Can be explicit → good for network analysis	Implicit, derived on the fly
Storage	Compact for discrete features	Large at high resolution; simple structure
Best for	Discrete features, cadastral/legal, networks	Continuous surfaces, fast arithmetic/overlay

Inter-conversion (both lose something): **Rasterization (V→R)** is relatively easy (points→cells, lines→cell sequences, polygons→cell zones) but introduces pixelation/stair-stepping. **Vectorization (R→V)** is much harder (polygonize contiguous same-value cells; may create noisy “speckle” polygons). **Dominant / winner-takes-all rule:** when a cell is covered by several vector features, assign the value of the feature covering the largest area (or whose centroid falls in the cell) – misleading for genuinely mixed cells (e.g. 40% residential/35% commercial/25% park all becoming “residential”).

2.11 Thematic Mapping & Choropleth Classification

A **choropleth map** shades/colours polygon features by an attribute’s value – best for rate/ratio/normalized attributes (literacy %, density, forest cover %). It requires **two decisions**: which attribute to map, and how to classify it.

Classification methods – know good-for / disadvantage of each

Method	How	Disadvantage
Equal Interval	Equal-width value ranges	Skewed data crowds one class
Pretty Breaks	Equal interval, rounded to neat numbers	May not match the data’s real distribution
Quantile (equal count)	Same number of features per class	Similar values can be split apart; wide ranges lumped together
Standard Deviation	Classes by distance from the mean (SD units)	Needs normally-distributed data; outliers skew the mean
Natural Breaks (Jenks)	Minimizes within-class variance	Dataset-specific; hard to compare across maps

Other thematic types: **graduated symbols** (size steps by class), **proportional symbols** (size directly proportional, continuous), **dot-distribution maps** (1 dot = fixed count), **isopleth/isoline maps** (equal-value lines over a continuous surface).

2.12 Practice & Review

Q1. [MC] Which vector structure stores a shared polygon boundary twice, once per polygon?

- (A) Arc-node
- (B) DIME
- (C) Spaghetti
- (D) Vertex dictionary

Answer: (C) Spaghetti has no topology, so each polygon is digitized independently, duplicating shared edges.

Q2. [MC] For computing NDVI pixel-by-pixel as fast as possible, the ideal band-interleaving format is:

- (A) BSQ
- (B) BIL
- (C) BIP
- (D) RLE

Answer: (C) BIP keeps all bands for one pixel contiguous, so NIR and Red for the same pixel need no extra file seeking.

Q3. [Short Answer] Classify: temperature in °C; rainfall in mm; “low/medium/high”; wind direction. *Answer:* Interval; Ratio; Ordinal; Cyclic.

Q4. [Short Answer] Why can interval data not be meaningfully multiplied or expressed as a ratio? *Answer:* No true zero – 0°C is not “no temperature,” so “twice as hot” is meaningless; only differences (subtraction) are valid.

Q5. [Essay] Explain the “dominant rule” for converting a mixed-class area to a single raster cell value, and when it misleads. *Answer:* The cell is assigned the value of whichever vector feature covers the largest proportion of its area (or whose centroid lies in the cell). It misleads when no single class truly dominates (e.g. three roughly-equal land uses in one cell) – the output silently discards the minority classes, which downstream analyses may then treat as if they didn’t exist there at all.

3 14, 18, 21 Aug: Spatial Data Collection

3.1 Overview

Data collection is one of the most **time-consuming and expensive** GIS tasks – typically **15–50% of total project cost**. Workflow cycle: **Planning** → **Preparation** → **Digitizing** → **Editing/Improvement** → **Evaluation** → back to Planning.

The Primary/Secondary × Vector/Raster 2×2 – classic table-completion question

	Raster	Vector
Primary	Digital RS images; digital aerial photos	Survey (total station); GPS measurements
Secondary	Scanned maps; scanned photos	Topographic surveys / digitized from maps

3.2 Remote Sensing (14 Aug)

Process (7 steps): Sunlight → Atmosphere → Earth features → Satellite/sensor → Antenna/receiver → Computer analyst → Application. Sensors detect **reflected** radiation (daytime, from sunlight) and/or **emitted** (thermal) radiation; at night, only emitted radiation is observed. Atmospheric effects (clouds, scattering) are the **dominant noise source**, lowering the Signal-to-Noise Ratio.

Spectral signatures (know the shapes):

- **Vegetation** – low reflectance in visible red (chlorophyll absorption), very high in **NIR** (leaf-cell scattering) – the sharp “red edge.”
- **Water** – strongly absorbs NIR and SWIR; very low overall reflectance; deep dips near 1.4 and 1.95 μm .
- **Soil** – gradual rise from visible to NIR; flatter curve, no red edge, no water-style dip. Dry soil reflects more than wet soil.

True vs. false colour composite: True colour maps sensor R/G/B to display R/G/B. False colour (e.g. NIR→Red, Red→Green, Green→Blue) makes healthy vegetation appear bright red.

3.2.1 Three (Four) Types of Resolution

Near-certain quiz item

Resolution	Definition	Example
Spatial	Ground area per pixel (ground sample distance)	1 km = low; 30 m = medium; 4 m = high; sub-metre = ultra-high
Spectral	Number/width of wavelength bands recorded	3 bands (R,G,IR) vs. many narrow hyperspectral bands
Temporal	Revisit frequency (how often the same spot is imaged)	Compare Landsat MSS-1976 vs. TM-1989 for the same city
(Radiometric)	Bit depth – number of brightness levels	8-bit vs. 16-bit

3.2.2 Pre-processing / Error Corrections

- **Geometric correction/georeferencing** – assign a coordinate system, correct sensor tilt & Earth curvature; use Ground Control Points (GCPs) from toposheets or GPS/DGPS.
- **Radiometric correction/normalization** – adjust for sensor differences, atmosphere, sun angle, season; convert DN to surface reflectance.
- **Atmospheric correction.**

Change-detection workflow: define study area & resolution → acquire multi-temporal imagery → preprocess & co-register → classify/compute indices → compare to quantify change.

3.3 Aerial Photography & Platforms (14/18 Aug)

Satellite RS and aerial photography (AP) are technically similar; they differ in **capturing and interpretation**. AP uses an analog optical camera (nose/underbelly mounted, low altitude 3000–9000 m), later rasterized by scanning the film.

Satellite vs. Aircraft vs. Drone – trade-off table

Platform	Coverage/Revisit	Spatial res.	Atmospheric noise
Satellite	Regional/global; consistent but limited temporal res.	Coarse → high (30 m → sub-m)	Highest – clouds, scattering; low SNR
Aircraft	Medium-scale; mission-scheduled, repeat = costly	High (often <1 m)	Moderate; geometric distortion from platform motion
Drone/UAV	Small-area; cheap, on-demand repeat	Very high (cm-scale)	Lowest – flies below clouds; highest SNR

Drone payloads: RGB camera (daylight only), multispectral (NDVI, limited resolution), thermal/IR (temperature, no geology), SAR (all-weather, lower resolution), LiDAR (direct 3-D terrain, costly).

3.4 LiDAR (18 Aug)

LiDAR = Light Detection And Ranging, an **active** sensor (emits its own laser, ≈ 1064 nm, NIR); works day or night. System = airborne LiDAR scanner + GPS/IMU on the platform + a **ground GPS base station** (corrects systematic satellite-clock/atmospheric error $\rightarrow$ cm-level XYZ per pulse).

Multiple returns per pulse

First return = highest surface hit (canopy top, roof) $\rightarrow$ used for canopy-height models, **DSM**. **Last return** = lowest hit (usually ground) $\rightarrow$ used for bare-earth **DEM/DTM**. Leaf-off deciduous canopy lets more pulses reach the ground than leaf-on or coniferous canopy.

DEM/DTM = bare-earth elevation. **DSM** = includes trees/buildings. **TIN** = a Delaunay-style triangulated surface built from the point cloud. **Error sources**: GPS error, IMU drift, timing errors, atmospheric effects, misclassified returns (roof mistaken for ground), multipath, vegetation penetration limits.

3.5 Radar/SAR & Sonar (18 Aug)

RADAR/SAR – active microwave; longer wavelengths penetrate clouds/rain, enabling monsoon and night imaging. **Polarization**: **Co-polarized (HH, VV)** – sensitive to surface roughness/geometry, strong returns from smooth surfaces and oriented man-made structures. **Cross-polarized (HV, VH)** – dominated by volume scattering (canopies, crops); good for vegetation structure/moisture.

Sonar/Bathymetry – acoustic pulse analogue of LiDAR for underwater terrain.

Sonar depth formula

$$\text{Depth} = \frac{\text{speed of sound in water} \times \text{two-way travel time}}{2}, \quad \text{speed} \approx 1500 \text{ m/s (varies with temperature, salinity)}$$

Error sources: sound-speed variability, multipath, wide-beam footprint averaging over rough terrain, rough-sea scattering, timing errors.

3.6 GPS/GNSS (21 Aug)

Three segments: **Space** (24+ satellites, 6 orbital planes, 4/plane, altitude 20,200 km, ~ 12 -hour orbit, 55° inclination, atomic clocks); **Control** (Master Control Station + monitor stations); **User** (receivers).

The core GPS equation

$$\text{RATE} \times \text{TIME} = \text{DISTANCE}$$

RATE = speed of light $\approx 300,000$ km/s. Each measured pseudo-range places you on a sphere around that satellite; intersecting ≥ 3 spheres (plus a 4th satellite to solve receiver-clock bias) gives (X, Y, Z) . **A 1 ms clock error $\Rightarrow$ ~ 300 km position error** – nanosecond-level timing is required for metre-level positioning.

Error sources: blocked signal (need ≥ 3 unobstructed satellites – fails in forests/urban canyons), **multipath** (signal reflects off buildings before arriving), satellite clock drift, ionospheric/tropospheric delay, ephemeris (orbit) error.

DGPS – Differential GPS

A **base station at a precisely known location** computes the difference between its measured and true position – this error vector (clock, atmosphere, ephemeris, multipath) is broadcast as corrections (often RTCM format) to nearby **rover** receivers, which subtract the common-mode error to reach **centimetre-level** accuracy instead of several metres. Base and rover must share the same area so the errors are common; correction can be real-time or post-processed.

3.7 Ground Surveying & Secondary Data Capture

Ground surveying: direct field measurement (points, distances, elevations, angles) via theodolite → **total station** (millimetre-level accuracy for both angle and distance).

Secondary raster capture: **digital scanning** (quality depends on source & scanner resolution), **rasterization** (vector→raster), **spatial interpolation** (estimate unknown cell values from weighted sample points – **IDW** simple distance-weighting vs. **Kriging** statistical/geostatistical), **resampling/grid aggregation** (change resolution, e.g. 10 m→30 m, via nearest-neighbour/bilinear/cubic).

3.8 Digitization & Its Errors (Secondary Vector Capture)

Digitization converts continuous lines into discrete points for computer storage: **manual (tablet)** digitizing (puck on an electromagnetic tablet), **heads-up (on-screen)** digitizing (trace over a scanned, georeferenced image), **automated vectorization** (scan → binary image → software traces lines).

Typical digitizing errors – overlaps heavily with the 4/8 Sept material

Undershoot (line stops short of its target, leaving a gap); **Overshoot/dangle** (line extends past the intersection); **Sliver polygons** (thin gap/overlap where two boundaries that should coincide don't quite match); **Invalid polygons** (fail to close); **sheet-edge mismatches** across adjacent map sheets.

Fixing errors in QGIS: the **Snapping tool** (forces new vertices to snap within a tolerance, eliminating gaps/overlaps); the **Topology Checker** (rules like “must not have gaps”/“must not overlap”/“must not have dangles”); the **Vertex tool** (move/add/delete vertices) followed by re-validation.

3.9 Data Transfer & Participatory Collection

Vector transfer: file-based (Shapefile, GeoPackage, GeoJSON, KML/KMZ, GML), database (PostGIS, SpatiaLite), web-based (**WFS** for vector, **WMS/WCS** for raster), GPS transfer (GPX, CSV). **VGI vs. crowdsourcing vs. citizen science:** VGI = individuals voluntarily contribute geographic data (OSM); crowdsourcing = many people via apps (flood/road reports); citizen science = public participation in scientific data collection (eBird).

3.10 Practice & Review

Q1. [MC] Which LiDAR return is used to build a bare-earth DEM?

- (A) First return
- (B) Last return
- (C) Median return
- (D) Maximum-intensity return

Answer: (B) The last return is typically the lowest surface hit (the ground), used for DEM/DTM; the first return (canopy/roof tops) builds the DSM instead.

Q2. [MC] A 1 millisecond GPS clock error translates to a position error of approximately:

- (A) 300 m
- (B) 3 km
- (C) 300 km
- (D) 30,000 km

Answer: (C) $0.001\text{ s} \times 300,000\text{ km/s} = 300\text{ km}$.

Q3. [Short Answer] Distinguish co-polarized from cross-polarized SAR sensitivity. *Answer:* Co-pol (HH, VV) is sensitive to surface roughness and geometry – strong returns from smooth surfaces and oriented man-made structures. Cross-pol (HV, VH) is dominated by volume scattering from complex 3-D targets like tree canopies – useful for vegetation structure and moisture.

Q4. [Short Answer] Why does DGPS need a base station at a *known* location? *Answer:* Because its true position is already known, any discrepancy between its measured and true position is entirely due to shared error sources (clock, ionosphere/troposphere, ephemeris, multipath); that error can then be broadcast as a correction to nearby rovers experiencing the same errors.

Q5. [Essay] A paper toposheet must be digitized into a clean vector layer. List the errors that typically occur and the QGIS tools/workflow to fix each. *Answer:* **Undershoot/overshoot** – fixed with the **Snapping tool** at an appropriate tolerance so new vertices lock onto existing ones. **Sliver polygons** – caught by running the **Topology Checker's** “must not overlap”/“must not have gaps” rules, then corrected with the **Vertex tool**. **Invalid (unclosed) polygons** – flagged by the same topology rules and closed by adding the missing vertex/arc. **Sheet-edge mismatches** – resolved via edge-matching (adjusting one sheet's boundary vertices to align with the neighbouring sheet). After each fix, re-run validation until the layer passes all topology rules.

4 4 Sept & 8 Sept: Data Cleaning, Topology and Generalization

4.1 Overview

This is a single slide deck (*Class 4*, split across two class days) with three big themes: (1) where errors enter a GIS dataset and how to fix them, (2) object relationships and topology, and (3) scale, generalization, and querying spatial data. Note: no transcript/summary exists for 8 Sept – that day’s content is reconstructed purely from its slide decks (Parts 2–3) and the classwork.

4.2 Primary Data Capture Errors

1. **Positional accuracy of GPS receivers** – satellite clock bias, satellite orbit (ephemeris) error, ionospheric/tropospheric delay, multipath, receiver clock bias, and the set of visible satellites (recall Section 3’s GPS/DGPS material – this is the same error taxonomy revisited).
2. **Device/instrument malfunction of electronic scanners: systematic noise** (regular striping/banding), **random noise** (salt-and-pepper speckles), **line dropouts** (missing scan lines, e.g. Landsat 7 SLC-off; fixed by gap-filling – averaging preceding/succeeding line values).
3. **Geometric distortion**: *Aerial* – altitude change rescales the image; attitude changes (**roll**, **pitch**, **yaw**) cause compression/expansion or frame rotation. *Satellite* – **cross-track scanner distortion** (pixel size grows off-nadir), **scan skew** (mirror actual vs. nominal velocity), and distortions from Earth rotation, spacecraft velocity, or altitude/pitch/roll/yaw variation, each producing a characteristic parallelogram/trapezoid shape.

4.3 Secondary Data Capture Errors (Digitizing)

Node errors – snapping (same vocabulary as Section 3)

Undershoot (line stops short) and **Overshoot** (line runs past) arise without a proper snap tolerance. Fixing node errors **validates connectivity** and **validates contiguity**.

Polygon errors: **Sliver polygons** (spurious thin polygons from two layers that should share an identical boundary but don’t); **dangling node** (unconnected line-end that should have closed a polygon); **pseudonode** (a redundant vertex splitting one logical line into two arcs – fixed with **split/unsplit**); **missing label point**; **line crossing with no node**.

4.4 Vectorization Without Post-Processing

Direct vectorization of a classified raster without cleanup produces **over-segmentation**, **spikes/sliver polygons**, **gaps between polygons**, **dangling polygons**, and **irregular boundaries**. Post-processing (cleaning, smoothing, editing) is essential – raw vectorization is never exam-ready output.

4.5 Georeferencing with Imprecise Control Points

Poorly-matched GCPs displace/distort the whole georeferenced output (an overall shift plus per-point positional error). Quality is checked via **RMS error**.

RMS error of GCP residuals

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N [(\Delta X_i)^2 + (\Delta Y_i)^2]}$$

Lower RMS = better fit; target RMS <1 pixel; well-distributed (not clustered) GCPs are essential.

4.6 Transformation, Rubber-Sheeting, Edge-Matching, Warping

Transformation = translation, rotation, differential scaling, skew, via a source-control-point → destination-control-point **displacement link**. **Rubber-sheeting** adjusts a source layer onto a target layer using displacement links and snapping. **Edge-matching** reconciles mismatched adjacent-sheet boundaries (an extension of the Section 3 “sheet-edge mismatch” problem). **Warping** applies many displacement links across a whole layer.

Practical link: the 8-Sept classwork

The 8-Sept QGIS exercise (two 1950s USGS topo sheets, *Mariposa* and *Yosemite*) is a real instance of this: setting a Project CRS, then choosing a **NAD27→NAD83→NAD83(HARN)** transformation pipeline for each raster before digitizing a common polygon.

4.7 Data Entry & Image Interpretation Errors

Missing/incorrect attribute values are caught by comparing against a master list / spelling check. **Resampling methods** (revisiting Section 3): **Nearest neighbour** (fastest, blocky, preserves categorical data), **Bilinear** (4-cell weighted average, smoother, unsuitable for categorical data), **Cubic convolution** (16-cell weighted average, smoothest, most compute).

4.8 Object Relationships & Topological Relationships

Spatial relationships (e.g. “is contained in,” “intersects”) vs. **relationships as attributes** (stored as data rather than computed geometrically). Three kinds of feature relationship: (1) those that **construct** complex objects from primitives (a line from its points); (2) those **computable from coordinates** (point-in-area, area overlap); (3) those **not computable from coordinates** (two line *geometries* crossing vs. the real highways actually intersecting at grade rather than via an overpass).

Pair	Relationships
Point–Point	<i>is within</i> (a distance); <i>is nearest to</i>
Point–Line	<i>on line</i> ; <i>is nearest to</i>
Point–Area	<i>is contained in</i> ; <i>on border of area</i>
Line–Line	<i>intersects</i> ; <i>crosses</i> ; <i>flows into</i>
Line–Area	<i>intersects</i> ; <i>borders</i>
Area–Area	<i>overlaps</i> ; <i>is within</i> ; <i>is adjacent to</i>

Relationships as attributes – worked examples: a stream network’s “flows-in” relationship can be stored either as (a) each link holding its downstream link’s ID, or (b) each link holding a downstream *point* ID and each point separately holding a downstream *link* ID. An “is contained in” relationship (wells → containing county) is computed once, stored as a new attribute, then aggregated (COUNT wells per county). **Object pairs:** attributes that belong to a *pair* of objects (distance, trade, commuter flow) are stored as a matrix or as a directed origin→destination link.

4.9 Topology: Definition, Properties, Advantages

Definitions to quote

ESRI: “A GIS topology is a set of **rules and behaviors** that model how points, lines, and polygons share **geometry**.” Longley: “Topology is the science and mathematics of **relationships** used to validate the geometry of vector entities.” In one line: **topology is a mathematics approach that defines unchangeable spatial relationships.**

Changes under stretching	Stays the same (topological)
Distance, Angles, Relative proximities	Adjacency; most relations (“contains,” “crosses”); object type (area stays area)

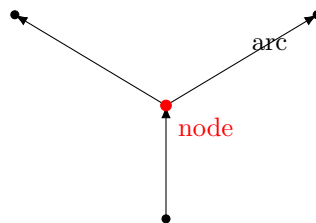
Advantages: efficient storage & quick processing of large datasets; enables spatial-model operations (network tracing, adjacency tests).

The 5 types of topological relationships – memorize

Connectivity (which features connect/intersect) · **Adjacency** (contiguity) · **Containment** (nestedness/insideness) · **Proximity** (nearness) · **Relative direction** (relative position).

4.10 Arc-Node Topology (ties directly to Section 2’s Arc-Node structure)

An **arc** is defined by a **from-node** and **to-node**; connected arcs share common node numbers, forming a **network**. Direction (from-node → to-node) is what makes **flow data** representable. **Left-right topology:** for each directed arc, the polygons on its left and right are its **adjacent** polygons.



Common node numbers link arcs into a network

Six named “topology types”: Arc-node (lines share endpoints), Polygon (areas share boundaries), Route (lines share segments), Region (areas can overlap), Node (lines share endpoints with points), Point events (points share vertices with lines).

Why topology matters – two classic slide examples

Shortest path pitfall: if junction topology is poorly defined (crossing segments treated as one line rather than properly noded), a shortest-path calculation gives the wrong answer.

Adjacency pitfall: two overlapped polygons must be cut into three distinct polygons to unambiguously define adjacency.

4.11 Map Scale, Generalization, Spatial Inference, Querying

Scale = map distance : ground distance, expressed as word statement, Representative Fraction (RF), or graphic scale. **Large scale** (1:12k–1:25k) = small area, high detail; **small scale** (1:250k+) = large area, coarse detail. Scale affects representation (a city is a point at small scale, a polygon at large scale) and even hydrologic simulation results (10 m vs. 100 m DEM of the same river).

10 map generalization operators

Simplification (weed out points), **Smoothing** (replace sharp forms), **Collapse** (area → point+line), **Aggregation** (many symbols → fewer new symbols), **Amalgamation** (several areas → one area), **Merging** (several lines → fewer lines), **Refinement** (complex pattern → a selection preserving its form), **Exaggeration** (enlarge to preserve a feature), **Enhancement** (alter symbol size/shape), **Displacement** (move to preserve visibility). Plus **Classification generalization** (reclassify attributes into fewer classes).

Spatial inference: **Interpolation** (predict between two known bounding values) vs. **Extrapolation** (predict beyond the known range, only one side known – less reliable).

Querying spatial data – 5 types: Cartographic/Map (find an ATM), Attribute-level (which road is longer?), Spatial (nearest hospital? does road X intersect a railway?), Spatio-Attribute (nearest ATM with > 4 machines), Topological (which survey numbers are affected by road widening? is IIIT-H in Hyderabad district?). Worked example: “Select all airports in India in seismic zone V” – a point-in-polygon spatial query.

4.12 Practice & Review

Q1. [MC] A vertex that appears twice at the same location, splitting one logical line into two arcs with no real topological meaning, is a:

- (A) Dangling node
- (B) Pseudonode
- (C) Sliver polygon
- (D) Overshoot

Answer: (B), removed via split/unsplit.

Q2. [MC] Which property survives a rubber-sheet stretch of a map?

- (A) Distance between two points
- (B) Angle between two lines
- (C) Adjacency between two polygons

(D) Relative proximity ranking

Answer: (C)

Q3. [Short Answer] Name the five types of topological relationships. *Answer:* Connectivity, Adjacency, Containment, Proximity, Relative direction.

Q4. [Essay] A district administration digitizes farm boundaries in a village. State two constraints that should apply and one that should not, with justification. *Answer:* Should apply: (1) two neighbouring farms cannot overlap; (2) a shared boundary segment belongs to at most two neighbouring farms. Should not apply: a rule forbidding a corner/node from belonging to more than two farms would be wrong – corners can legitimately be shared by three or more farms meeting at a point.

5 15 Sept: Vector Data Analysis

5.1 Overview

Vector data represents a simplified version of the world using points, lines, and polygons. Data enters a GIS via **selection** (which real-world features to include) and **representation** (point/line/polygon geometry, driven by scale – see Section 4). The heart of this lecture is **overlay operations**, grounded in **Boolean set logic** and computational geometry.

5.2 Boolean Logic in GIS

AND ($\cap$, intersection: in both A and B), **OR** ($\cup$, union: in either), **NOT** ($\neg$, difference: in A but not B), **XOR** (in exactly one). Bracket order matters: (A AND B) OR C $\neq$ A AND (B OR C).

5.3 Classification of Vector Overlay Operations

	Point	Line	Polygon
Points	Points coincide	Point in line	Point in polygon
Lines	Point in line	Line intersection	Line in polygon
Polygons	Point in polygon	Line in polygon	Polygon overlay

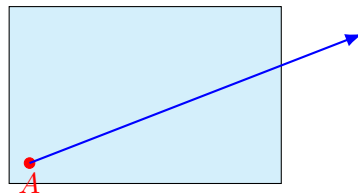
5.4 Point-in-Polygon: Ray-Casting / Odd-Even Rule

The rule

Cast a ray from the query point A in any fixed direction; count edge crossings. **Odd** = inside; **even** = outside. For line-intersection (needed to count crossings), given $y = a_1 + b_1x$ and $y = a_2 + b_2x$:

$$x_i = -\frac{a_1 - a_2}{b_1 - b_2}, \quad y_i = a_1 + b_1x_i$$

then check x_i falls within *both* segments' valid range.



1 crossing (odd) $\Rightarrow A$ is inside

5.5 Worked Overlay Examples & Overlay Patterns

Point in polygon (restaurants in wards), **line in polygon** (roads split at soil-type boundaries), **polygon overlay** (watershed $\times$ district). Other patterns: **Query** (AND across two coverages), **Reclassify** (collapse categories before overlay), **Coverage rebuilding** (clip + erase), **Overlay ranking** (look up a combined rank from two ranked layers), **Connectivity** (rezonify a buffer using connectedness). **Spatial Join** attaches attributes from multiple layers to one feature based purely on spatial relationship.

5.6 Clip vs. Intersect vs. Union vs. Difference – THE Comparison Table

The single most re-tested distinction in this lecture

Operation	Output geometry	Attribute handling
Clip	Only the part of A inside B (cookie-cutter)	Only A's attributes carry over
Intersect	Only the overlapping region of A and B	Both A's and B's attributes appear
Union	All of A and all of B	Both attribute tables, everywhere applicable
Difference/Erase	All of layer 1 not overlapping layer 2	Only layer 1's attributes remain
Symmetrical Difference	All areas except where they overlap	Attributes of whichever input contributed

Memory hook: Clip = “keep A's shape, throw away B's data.” Intersect = “keep the overlap, keep both tables.” Union = “keep everything, keep both tables.”

5.7 Data Manipulation & Buffering

Reclassify (collapse fine categories to coarse) → **Dissolve & merge** (remove now-redundant internal boundaries between adjacent same-class polygons). **Buffering**: points (circles, may merge), lines (a constant-width ribbon), areas (a zone extending outward/inward).

Tool	What it does	Best use case
Buffer	Zone of specified distance around features	Schools within 1 km of a road
Clip	Extract features overlapping a clip polygon	Restrict to a study-area boundary
Intersect	Geometric intersection common to all inputs	Overlapping zoning + flood zones
Union	Combine layers, keep all features/attributes	Overlay admin boundaries with ecological zones
Erase	Remove areas overlapping the erase layer	Exclude protected areas from development
Merge	Combine same-geometry-type datasets into one	Combine road networks from different districts
Dissolve	Aggregate by shared attribute, remove internal boundaries	Districts into provinces
Spatial Join	Join attributes by spatial relationship	Population from polygons to points
Select by Location	Select by spatial relation to another layer	Buildings within a flood zone

5.8 Case Study: Flood-Exposed Villages, Schools & Roads

Buffer river → **Intersect**(village, buffer) → **Select** schools inside → **Intersect**(road, flood zone) → **Erase** flood zone from usable land → **Union**(flood zone, land-use) → **Dissolve** into one zone → Merge multi-agency road layers.

5.9 Practice & Review

Q1. [MC] You overlay a school (point) layer with a flood-zone (polygon) layer using **Clip**. What happens to the flood zone's attributes?

- (A) Appear on every clipped point
- (B) Discarded – only the school's own attributes remain
- (C) Replace the school's attributes
- (D) Averaged with the school's attributes

Answer: (B)

Q2. [Short Answer] You need every part of two overlapping zoning layers preserved, with both layers' attributes everywhere. Which overlay, and why not Clip? *Answer: Union* – Clip would discard the second layer's attributes and restrict geometry to the clip feature; Union keeps the full extent and both attribute tables.

Q3. [Essay] Design a workflow to find residential parcels within 500 m of a river *and* inside a flood-risk polygon. *Answer: Buffer* the river 500 m → **Intersect** the buffer with the flood-risk polygon (AND logic) → **Select by Location** the residential-parcel layer against that combined zone → optionally **Dissolve** by parcel ID if overlay fragmented any parcel.

6 18 Sept: Raster Data Analysis – Part 1

6.1 Overview

Raster GIS is efficient for **overlaying** many thematic layers, allows **mathematical operations** across layers cell-by-cell, and provides a natural frame for **surface modelling** and **3D visualization**.

6.2 Raster Functions – The Big Picture

Map Algebra (mathematical operations across raster layers) and **Neighbourhood functions** (Euclidean, Metric, Topological – e.g. flow direction/accumulation).

Map Algebra operator families

Arithmetic: + - * / ^ **Boolean:** True/False **Comparison:** < ≤ = > ≥ **Logical:** contained-within-sets **Combinatorial:** overlay of different grids **Accumulative:** sum/divide/multiply groups **Assignment:** assigning values during operations.

6.3 Raster-Based Overlay

Simple addition (cell-by-cell sum of scored grids); **Boolean combine** (geometric phase – common grid framework; then attribute phase – threshold + AND/OR combination rule).

Worked example: Environmental Contamination Risk Index

Normalize each raw layer to 0–1 (divide by its max), then weight and sum:

$$\text{Risk Index} = 0.4 \cdot \text{HMC}_{\text{norm}} + 0.3 \cdot \text{Perm}_{\text{norm}} + 0.3 \cdot \text{Slope}_{\text{norm}}$$

This normalize → weight → sum pattern is the template for almost any weighted multi-criteria raster overlay question.

6.4 Overlay: The 4 Basic Rules

Enumeration (every unique combination preserved) · **Dominance** (one value wins, e.g. max) · **Contributory** (each value adds, e.g. summing 0/1 layers) · **Interaction** (a pair of values jointly determines the output via a lookup of all value-pairs).

6.5 Cell/Pixel/Grid Operation Classification

The single most-repeated classification question in this course

Type	Depends on	Example
Local	Only the cell's own value(s) across aligned grids	<code>new_map = old_map_1+old_map_2;</code> single-cell reclassification
Focal	Cell x , $f(x)$, and its neighbourhood $n(x)$	3×3 mean filter; buildings within 5 km of a fire station
Zonal	All cells in the same zone	Average slope per watershed
Global (Incremental)	The entire raster grid	Cost-distance from a source; least-cost path

6.6 Local Functions & Reclassification

Generic form $U = f(X_1, X_2, \dots)$: single-raster stats (max/min/range/sum/mean/median/SD; majority/minority for categorical); multi-raster ops – **mean**, **merge** (fills NODATA: `if X1==NODATA then U=X2 else U=X1`), **max/min**; multiply by constant vs. by another grid; **combine** (unique output per unique input combination).

Reclassification: Arbitrary (lookup table, `con(ingrid==1,5,...)`); **Boolean** (two-input coding-scheme lookup, `if X1==a1 and X2==b1 then U=c1 ...else NODATA`); **Statistical/Slice** (**EQAREA** – equal cell-count zones vs. **EQINTERVAL** – equal-width value ranges). **Advantages**: simplification, unique category creation, ranking.

6.7 Practice & Review

Q1. [MC] “Deriving a surface-waterbodies map by reclassifying lakes/ponds/rivers into one class” is:

- (A) Focal
- (B) Zonal
- (C) Local
- (D) Global

Answer: (C) – a per-cell lookup-table reclassification, no neighbours or whole-grid dependency.

Q2. [Short Answer] Distinguish the Dominance rule from the Contributory rule, with an example each. *Answer:* Dominance picks one winning value (e.g. **max** of bid rasters); Contributory lets every value add (e.g. summing binary hazard-presence layers into a hazard count).

Q3. [Essay] Why normalize raster layers to 0–1 before a weighted overlay? *Answer:* Without normalizing, whichever layer has the largest raw numeric range dominates the weighted sum regardless of its assigned importance weight.

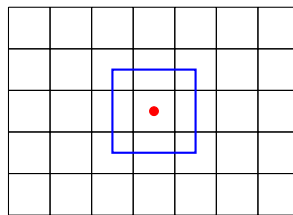
7 19 Sept: Raster Data Analysis – Part 2

7.1 Overview

Continuation of Local/Focal/Zonal/Global (Section 6): this lecture goes deep on **Focal**, **Zonal**, and **Global** operations.

7.2 Focal Functions (Neighbourhood)

Relate a cell to its neighbours (chosen by distance and/or direction); result written to the focal (centre) cell; window slides across the grid. **3 required parameters**: target location(s), neighbourhood specification, function to apply. **Shapes**: Rectangle, Circle (isotropic), Annulus (ring), Wedge/Pie (directional).



A 3×3 rectangular kernel centred on the focal cell

7.3 Neighbourhood Analysis / Spatial Filtering

Low-pass (mean) filter: kernel of $1/9$ s (3×3), smooths; weights must sum to 1.0. **Median filter**: robust to outliers. **High-pass (edge-detection) filter**: negative-weight kernel (e.g. centre $+9$, neighbours -1), exaggerates local detail. **Standard-deviation (texture) filter**: high SD = high texture/variability (an edge); low SD = homogeneous.

Convolution $\approx$ Focal, generalized

The CNN convolution operation is explicitly an extension of the focal operation: a kernel slides over a grid exactly like a focal filter, except the weights are learned rather than fixed.

Worked example: how many residential buildings lie within 5 km of a fire station? Target = fire station; Neighbourhood = 5 km circular kernel; Function = count.

7.4 Zonal Functions

Operate on a **zone** (group of cells sharing a value/region), which may be **contiguous** or **non-contiguous**. **Do not change zone boundaries**; only change the attribute value per zone. Needs **two grids**: a zone grid and a value grid.

Worked example: Zonal Area

1 cell = 250,000 m²; every cell in a 7-cell zone gets the value $7 \times 0.25 \text{ km}^2 = 1.75 \text{ km}^2$ (the zone's total area).

Applications: landscape-ecology geometry measurement; comparing statistics between rasters (mean *slope* per *watershed*).

7.5 Global (Incremental) Functions

Output depends on the **entire** grid. Examples: distance from a source to every other cell, optionally weighted by friction → **cost-distance surfaces**. Uses: diffusion modelling, shortest-path, site selection, visibility, connectivity/fragmentation.

Proximity: Euclidean distance (straight-line) vs. **cost distance** (friction-weighted); the **least-cost path** is the minimum-cost route, often different from the shortest Euclidean path. **Barrier effects**: no barrier = circular rings; **absolute barrier** = rings detour entirely around it; **partial barrier (friction)** = rings compress/slow through it but still cross.

Spread functions: a land-use raster (Cropland, Road, Rangeland, Quarry) converted cell-by-cell into a travel-time raster using domain-known speeds – this friction surface feeds cost-distance/least-cost-path. **Generalized raster buffering**: vary the buffer distance by an accessibility layer (e.g. “reachable within 5 min” vs. “10 min”) instead of a fixed radius.

7.6 Practice & Review

Q1. [MC] A 3×3 mean filter kernel must have weights that:

- (A) Sum to 0
- (B) Sum to 1.0
- (C) Are all negative
- (D) Sum to the kernel size squared

Answer: (B)

Q2. [Short Answer] Why does a zonal function require two rasters, and what happens to zone boundaries? *Answer:* One raster defines the zones, one supplies the values to summarise; boundaries never change, only the attribute value assigned within each zone.

Q3. [Essay] Combine a focal and a zonal operation into one realistic municipal-planning workflow. *Answer:* Focal-count hospitals within 5 km per cell → zonal-mean income per administrative ward (broadcast back to every cell in the ward) → combine both via a local weighted-sum to score candidate ambulance-station sites.

8 Previous-Year Question Bank (All 4 Papers)

Sources

Quiz 1 – Set A (Ref/SI_quiz1_2024.pdf, “Monsoon 2024,” dated 28 Aug 2023 on the header); **Quiz 1 – actual 2026 paper** (Ref/SI Quiz 1.pdf, administered 1 Sept 2026 – your own recent exam, very high signal); **Mid-Sem 2024** (Ref/SI_mid1_2024.pdf, 21 Sept 2024); **Mid-Sem 2025** (Ref/GS2.401 Midsem.pdf, 25 Sept 2025). Questions requiring **map projections or the Geoid** are excluded – not covered in any of the 12 classes to date. Everything else is adapted (reworded, not verbatim) into practice form below, cross-referenced to the section that covers it.

8.1 From the Quiz 1 Papers

Q1. [MC] Which is the first recorded example of spatial analysis in GIS?

- (A) Anthrax outbreak in Sverdlovsk
- (B) Great Hanshin earthquake
- (C) Cholera outbreak in London
- (D) COVID-19 pandemic

Answer: (C) – John Snow’s 1854 map (Section 1).

Q2. [MC] Which of the following is *not* a vector analysis function?

- (A) Proximity analysis
- (B) Buffer analysis
- (C) Spatial overlay operations
- (D) (trick option naming something raster-only)

Answer: Proximity, buffer, and overlay are all core vector functions (Sections 5, 7); watch for an option describing a *raster-only* operation (e.g. “focal filtering”) being the odd one out.

Q3. [MC] Finding the nearest branch of *my own* bank from IIIT-H campus is what type of query?

- (A) Attribute Query
- (B) Spatial Query
- (C) Spatio-attribute query
- (D) Spatio-temporal query

Answer: (C) *Spatio-attribute* – “nearest” is spatial, but “in which I have an account” is an attribute filter first; this combination is exactly the Spatio-Attribute query type from Section 4.

Q4. [Multi-select] Which of the following can be represented as spatial objects in *either* Vector or Raster models: disease occurrence; pest infestation in fields; a rice mill; a snow avalanche; over-the-road railway intersections? *Answer:* All of them, in principle – discrete point-like phenomena (rice mill, a railway intersection) fit vector; diffuse/continuous phenomena (disease occurrence density, pest infestation extent, avalanche runout) fit raster; several (avalanche extent, infestation extent) can be modelled either way depending on whether a crisp boundary or a continuous risk surface is more useful (Section 2’s vector-vs-raster trade-off).

- Q5. [Short Answer]** What are “S,” “G,” “I” generally understood to stand for across the course’s framing of GIS (Systems/Science/Services)? *Answer:* See Section 1: **GISystems** = tools/storage/formats; **GIScience** = analysis/patterns/modelling; **GIServices** = data/capability delivered to users. (“Structure” and “Scenarios” are distractors, not standard GIS terminology.)
- Q6. [Short Answer]** A neighbourhood land-use map has more polygons than land-use classes, including several very small ones. What are these small polygons called, and what likely caused them? *Answer:* **Sliver polygons** (Section 4) – caused by two boundaries that should coincide (e.g. two adjacent land-use layers, or a layer digitized twice) not matching exactly, leaving a thin spurious polygon after overlay/union.
- Q7. [Short Answer]** Which vector data model stores coordinate strings independently with no internal topological structure? *Answer:* The **spaghetti model** (Section 2).
- Q8. [Short Answer]** Which instrument measures both angles and distances to millimetre-level accuracy? *Answer:* The **total station** (Section 3).
- Q9. [Short Answer]** Which technique obtains information about an object’s properties without physical contact? *Answer:* **Remote sensing** (Section 3).
- Q10. [Short Answer]** Which process converts continuous line graphics into a series of discrete points for computer storage? *Answer:* **Digitization** (Section 3).
- Q11. [Short Answer]** Which surface model depicts *complete* topography including both natural and man-made features (as opposed to bare earth)? *Answer:* **DSM (Digital Surface Model)** (Section 3) – includes buildings/canopy, unlike the bare-earth DEM/DTM.
- Q12. [Short Answer]** What is the measurement/mapping of underwater terrain and water depth called? *Answer:* **Bathymetry** (Section 3, via sonar).
- Q13. [Short Answer]** An ecologist has point samples of individual plants and wants a continuous surface showing spatial *density*. Which technique? *Answer:* **Spatial interpolation** (e.g. Kernel Density / IDW / Kriging-style surface generation, Section 3) – converting discrete point samples into a continuous raster surface.
- Q14. [Short Answer]** Which technique obtains highly precise ground control points for geometrically correcting high-resolution imagery? *Answer:* **DGPS** survey (Section 3), or an RTK-style survey providing sub-metre/cm GCPs.
- Q15. [Essay]** A city pollution survey gives point readings across an imaginary city boundary. How would you represent the spatial variation using GIS, and in what pictorial format? *Answer:* Treat the readings as sample points, then generate a **continuous raster surface via spatial interpolation** (IDW or Kriging, Section 3), and visualize it as an **isopleth/isoline map** or a pseudo-colour raster (Section 2) – a choropleth over the city boundary is *not* ideal here since pollution is a continuous field, not a per-polygon rate.
- Q16. [Short Answer]** Classify: Phone numbers; Rainfall (mm); Population; Height of trees. *Answer:* Phone numbers – **Nominal** (an identifier, not a true quantity, despite being digits); Rainfall – **Ratio**; Population – **Ratio**; Height of trees – **Ratio** (Section 2).

- Q17. [Short Answer]** List two problems that arise when converting a paper map into a digital map.
Answer: The source paper stretches/shrinks with humidity/age causing point drift; the source map's own drawing errors get baked in; digitizing itself introduces undershoot/overshoot/sliver errors; adjacent sheets may mismatch at their shared edge (Sections 3, 4).
- Q18. [Essay – mixed pixel]** A raster cell straddles Tree/Airport/House classes (a mixed-pixel situation). Identify two methods to assign a single class to the pixel and sketch the resulting classified grid for each.
Answer: (1) **Dominant/winner-takes-all** – assign the class covering the largest area within the cell (Section 2). (2) **Centre/nearest-neighbour rule** – assign the class located at (or nearest to) the cell's centre point, regardless of area coverage (the raster-resampling “nearest neighbour” idea, Section 4). The two methods can disagree: a small class sitting exactly at the cell centre wins under rule (2) even if a larger class covers more area (rule 1 would pick that larger class instead).
- Q19. [Essay – RLE]** Represent a given raster image (rows of mostly-uniform values with a blob of a different value) using Run-Length Encoding.
Answer: Scan row by row; for each row emit (count, value) pairs for consecutive runs of the same value, e.g. a row 0 0 0 1 1 0 0 0 becomes (3,0) (2,1) (3,0) (Section 2). Rows that are entirely one value compress to a single pair; rows crossing the blob's boundary produce 3 pairs.
- Q20. [Essay – topology figure]** Given a figure with labelled nodes (N1–N4), links (L1–L6), and polygons (A–E), write out the polygon topology, node topology, link topology, and link coordinate table.
Answer: This is a direct **arc-node topology** construction exercise (Sections 2, 4). Build three linked tables: **Node table** (Node ID, Easting, Northing) for each labelled node; **Link/Arc table** (Link ID, From-Node, To-Node, Left-Polygon, Right-Polygon) for each labelled link, reading the left/right polygon off the figure relative to the arc's drawn direction; **Polygon table** (Polygon ID, ordered list of bounding Link IDs) for each labelled polygon (including the unbounded “outside” polygon, often labelled E). Finally, a **Link coordinates table** lists the vertex-by-vertex (X, Y) string for each link (its full geometry, not just endpoints).

8.2 From the Mid-Semester Papers (2024 & 2025) – Multi-Select Style

- Q1. [Multi-select]** Select all valid statements about vector data structures:

- (a) The spaghetti model preserves topological relationships between spatial objects.
- (b) Polygon-arc topology inherently makes sure no two polygons overlap.
- (c) Arc-polygon topology captures the adjacency relationship between two polygons.
- (d) A vertex dictionary stores topological relationships by avoiding duplication of vertices.

Answer: (c) only. (a) false – spaghetti has *no* topology at all (Section 2). (b) is a trap: topology *validates*/flags overlaps via rules, it doesn't automatically forbid bad input geometry. (c) true – left-right/arc-polygon topology is exactly how adjacency is derived (Section 4). (d) false – a vertex dictionary removes coordinate *duplication* but still carries **no topology** (Section 2); don't confuse “no duplicate vertices” with “has topology.”

- Q2. [Multi-select]** Which represent topological properties (survive a stretch/distortion)?

- (a) Campus area is exactly 66 acres
- (b) Road A is connected to the entry gate

- (c) The canteen is south-east of the football ground
- (d) A new construction site exists inside the campus

Answer: (b). Area (a) is metric, changes under stretching. Directional/relative-position (c) is excluded from the slide’s “stays the same” list (only adjacency and most contains/crosses relations are guaranteed). (d) is a plain fact, not a relationship.

Q3. [Multi-select] Digitizing agricultural farm boundaries: which topological constraint would you *not* enforce?

- (a) Two neighbouring farms can overlap
- (b) Two neighbouring farms cannot overlap
- (c) A boundary segment belongs to at most two neighbouring farms
- (d) Corners may belong to more than two neighbouring farms

Answer: (a) – overlap is exactly what you enforce *against*; (b), (c), (d) are all legitimate rules (Section 4).

Q4. [Multi-select] For a metro-network map: select all valid statements.

- (a) The metro map is topologically equivalent to the actual network
- (b) It correctly represents true distance between stations
- (c) Connectedness is often misrepresented
- (d) A curved “S”-shaped line on the ground can be a straight line on the map

Answer: (a) and (d) – a metro diagram is the classic example of pure topology without metric accuracy: it preserves connectivity/order and freely straightens curves *because* distance/shape are non-topological, while connectivity is preserved by design (Section 4).

Q5. [Multi-select] Select all valid statements for a raster model of spatial data:

- (a) A raster model better describes discrete geographic objects
- (b) A focal operation may use grid-cell values from a 3×3 neighbourhood
- (c) A raster map with a “bigger numerical value” inherently provides more spatial detail
- (d) Temperature, precipitation, and elevation are better represented by a raster model

Answer: (b) and (d) – vector better describes discrete objects (a is false); cell *value* magnitude has nothing to do with *resolution*/detail (c is false, Sections 2, 6).

Q6. [Multi-select] For Sentinel-2 (10 m) vs. LISS-IV (5.8 m) data, select the correct statements:

- (a) Sentinel data at 1:50K looks higher-resolution than at 1:100K
- (b) LISS-IV spatial resolution is higher than Sentinel’s
- (c) At 1:100K both give the same level of detail
- (d) LISS-IV can be down-sampled to ~ 10 m using a focal operation

Answer: (a), (b), (d). 5.8 m is a finer (higher) spatial resolution than 10 m, so (b) is true. Visualizing at a larger map scale (smaller denominator, 1:50K) reveals more of a raster’s native detail than a smaller scale (1:100K), so (a) is true. (c) is false – resolution differences persist regardless of display scale, they just become less *visible* at coarse display scales, which is not the same as “same level of detail.” (d) is true – a focal mean/aggregate operation is exactly how you coarsen 5.8 m data toward ~10 m (Sections 3, 7).

Q7. [Multi-select] Which can be represented as *both* vector and raster?

- (a) Roads with widths
- (b) A buffer of air pollution around a road
- (c) National forests
- (d) Property owned by individuals/organisations

Answer: All can, in principle, but the “natural fit” differs – roads-with-width and property parcels are naturally *vector* (discrete, precise boundaries) but can be rasterized; a pollution buffer and a forest extent are naturally *raster*-friendly (continuous/diffuse) but can be vectorized – this question tests that the same real-world object can cross the vector/raster line depending on the analysis need (Section 2).

8.3 From the Mid-Semester Papers – Scenario / Design Questions

Q1. [Short Answer] State the overlay type for: “district-wise (polygon) count of health-care facilities (points).” *Answer:* **Point-in-polygon overlay**, aggregated by `COUNT(...)` `GROUP BY district` (Sections 4, 5).

Q2. [Short Answer] State the overlay type for: “road segments (line) passing through forest areas (polygon).” *Answer:* **Line-in-polygon overlay** (Section 5).

Q3. [Short Answer] Classify: (a) deriving a surface-waterbodies map by reclassifying lakes/rivers into one class; (b) creating a shortest-distance map from two source points. *Answer:* (a) **Local**; (b) **Global** (per the slide definitions – computing distance-from-source across an entire grid is the textbook Global/incremental case; if your rubric expects “focal,” flag this terminology edge case to your TA, Section 6).

Q4. [Essay] Give a real-world example combining both **focal** and **zonal** operations. *Answer:* See the ambulance-siting example at the end of Section 7 (hospital-count focal raster + mean-income zonal raster, combined via local map algebra).

Q5. [Essay] After detecting waterbodies via NDWI + thresholding, spurious tiny (single-pixel) polygons appear. Cause, fix, and an alternate index? *Answer:* **Cause:** sensor/random noise (speckle) near the threshold boundary, or mixed pixels – the sliver-polygon/noise problem of Section 4. **Fix:** a **focal majority filter** before polygonizing, plus a minimum-area cleanup pass. **Alternate index:** NDBI/MNDWI, or cross-checking with a slope/elevation raster via a local Boolean AND.

Q6. [Essay] Argue vector sufficiency/insufficiency for (a) house footprints, (b) population density across zoned urban areas. *Answer:* (a) **Vector sufficient** – discrete, sharply-bounded objects. (b) **Vector insufficient alone** – population density is a continuous field varying within a zone; a raster (or a zonal operation distributing a zone’s total across cells) captures that intra-zone variation that vector zoning discards (Sections 2, 7).

- Q7. [Essay]** List errors in *manual* digitization of maps. *Answer:* Node errors (under-shoot/overshoot), pseudonodes, dangling nodes, sliver polygons, missing label points, unnoded line crossings, source-map stretch/shrink, sheet-edge mismatches (Sections 3, 4).
- Q8. [Essay]** Bank branches (points) over 500 sq. km must be checked so no two are within 10 km of each other. *Answer:* **Buffer** every branch by 5 km → find overlapping buffers via **Intersect/Select by Location** → flag one branch per overlapping pair as a relocation candidate; equivalently, run a **proximity/Euclidean-distance** global function and select pairs <10 km apart (Sections 5, 7).
- Q9. [Essay]** Urban-growth model uses nearness to existing urban area and closeness to roads as key parameters – which mode of analysis? *Answer:* A **weighted raster overlay** of two **proximity/cost-distance** surfaces (distance-to-urban-area, distance-to-road), normalized and combined (Sections 6, 7) – exactly the multi-criteria weighted-sum pattern.
- Q10. [Essay – HYDRA buffer-zone scenario]** A water body’s buffer is 9 m (<10 Ha spread) or 30 m (≥10 Ha), from an old 1:25,000 paper toposheet FTL boundary plus a 10 m classified satellite water mask (3 recent high-rainfall years). Design a system to (A) tell the public whether a plot lies in the prohibited zone, and (B) count existing buildings inside it. *Answer:* **Georeference** the scanned toposheet into the satellite mask’s projection using GCPs (Section 4); **vectorize** the FTL boundary and each year’s water polygon, **Union** the 3 years to get the conservative max water spread, and compute area to pick 9 m vs. 30 m. **(A): Buffer** each water body by the right distance; a queried plot runs a contained-in spatial query against that buffer layer. **(B): Intersect/Select by Location** the building-footprint layer against the buffer, then COUNT.
- Q11. [Essay – pipeline scenario]** Centerline road data (width 8–48 m, duplicated names) must become (A) a pipeline offset 1 m from the left boundary, (B) a junction/turn dataset with type attributes. *Answer:* First convert the spaghetti-like centerline data into proper **arc-node topology** (snapping endpoints into shared nodes, resolving duplicate names via unique segment IDs). **(A):** offset the centerline by half its width to get the road boundary, then a further 1 m offset for the pipeline, storing diameter as an attribute. **(B):** every node where ≥3 arcs meet (or two meet at an angle) is a candidate junction/turn – extract as points and classify by connecting-arc count/angle, using **connectivity** (Section 4).
- Q12. [Essay]** A Mumbai–Hyderabad highway expands from 4 to 8 lanes. Given highway line data and a LULC raster, estimate impacted area per LULC class and propose flyover/land-acquisition strategy. *Answer:* **Buffer** the highway centerline by the width increase (a local widening buffer) → **overlay** (raster Boolean combine or vector Intersect, depending on LULC’s format) the buffer against the LULC layer → **zonal** statistics to tabulate affected area per LULC class. For flyovers/acquisition: prioritise segments where the buffer crosses *built-up* LULC (highest acquisition cost/impact) for flyovers (elevated, avoiding ground acquisition) and segments crossing low-value LULC (agriculture/barren) for at-grade widening with land acquisition (Sections 5, 6).

9 Master Cheat Sheet

Numbers to memorize

GPS: 24+ satellites, 6 planes, 20,200 km altitude, ~12 h orbit, 55° inclination.
 RATE×TIME=DISTANCE, RATE≈300,000 km/s. 1 ms clock error ⇒ ~300 km error.
 LiDAR wavelength ≈1064 nm. Sonar speed ≈1500 m/s, depth = speed×two-way-time/2.
 Data collection = 15–50% of project cost. RMS target <1 pixel.

Key formulas

Point-in-polygon: odd # crossings = inside.

Line intersection: $x_i = -\frac{a_1 - a_2}{b_1 - b_2}$, $y_i = a_1 + b_1 x_i$.

RMS: $\sqrt{\frac{1}{N} \sum (\Delta X_i^2 + \Delta Y_i^2)}$.

NDVI: (NIR−Red)/(NIR+Red) – best stored as **BIP**.

Mean 3×3 kernel: all 1/9 (sum to 1.0). **Edge kernel:** centre +9, 8 neighbours −1.

Weighted overlay: normalize to [0, 1] → × weight → sum.

Zonal area: every cell in a zone → (# cells)×(area/cell).

	Vector structures	Vector overlay
No topology	Spaghetti, Vertex dictionary	Clip (A's attrs only)
Has topology	DIME, Arc-node	Intersect (both attrs, overlap only) Union (both attrs, everything)

	Raster storage	Raster operation type
Per-band	BSQ	Local – one cell
Per-line	BIL	Focal – kernel/window
Per-pixel	BIP (best for NDVI)	Zonal – named zone, 2 grids
Compressed	RLE, Quadtree	Global – whole grid

Memorize as lists

5 Topological relationships: Connectivity, Adjacency, Containment, Proximity, Relative Direction.

4 Raster overlay rules: Enumeration, Dominance, Contributory, Interaction.

10 Generalization operators: Simplification, Smoothing, Collapse, Aggregation, Amalgamation, Merging, Refinement, Exaggeration, Enhancement, Displacement.

3(+1) Resolutions: Spatial, Spectral, Temporal, (Radiometric).

5 Choropleth classification methods: Equal Interval, Pretty Breaks, Quantile, Standard Deviation, Natural Breaks (Jenks).

10 Rapid Self-Test (Whole Course)

- Q1.** What made John Snow’s cholera map the first spatial analysis? *[Mapped deaths geographically, revealing a cluster around one contaminated pump, disproving the airborne theory.]*
- Q2.** Name the five eras of geospatial technology evolution. *[Field/analog → sensing/digital → digital maps web → Geo Services → GeoAI.]*
- Q3.** Which two vector structures have no topology, and which two do? *[Spaghetti & vertex dictionary: no. DIME & arc-node: yes.]*
- Q4.** What three things does DIME store per segment? *[Segment name; from/to node IDs; left/right polygon IDs.]*
- Q5.** Which measurement scale has no true zero? *[Interval.]*
- Q6.** Best band-interleaving format for computing NDVI, and why? *[BIP – all bands for one pixel are contiguous.]*
- Q7.** Which compression method suits a classified low-res raster but fails on a DEM? *[RLE (run-length encoding).]*
- Q8.** State the “dominant rule” for mixed-pixel classification. *[Assign the class covering the largest area / at the cell centroid.]*
- Q9.** Name the three (four) types of remote-sensing resolution. *[Spatial, spectral, temporal, (radiometric).]*
- Q10.** Distinguish first vs. last LiDAR return. *[First = canopy/roof top (DSM); last = ground (DEM/DTM).]*
- Q11.** DGPS core mechanism? *[A known-location base station computes its own error and broadcasts it as a correction to nearby rovers.]*
- Q12.** Name four typical manual-digitizing errors. *[Undershoot, overshoot/dangle, sliver polygon, invalid/unclosed polygon.]*
- Q13.** What is a pseudonode, and which tool removes it? *[A redundant same-feature vertex splitting one line into two arcs; removed via Split/Unsplit.]*
- Q14.** Give the point-in-polygon odd-even rule. *[Odd # of ray-boundary crossings = inside; even = outside.]*
- Q15.** Which overlay operation discards the “cookie-cutter” layer’s own attributes? *[Clip.]*
- Q16.** Which raster operation type does NOT change zone boundaries, only attribute values? *[Zonal.]*
- Q17.** Give the generic form of a local raster function. *[$U = f(X_1, X_2, \dots)$.]*
- Q18.** Interpolation vs. Extrapolation? *[Interpolation fills gaps between two known bounding values; extrapolation predicts beyond the known range, only one side known.]*
- Q19.** Which map-generalization operator replaces an area object with point+line objects? *[Collapse.]*

- Q20.** Name the 3 required parameters of any focal function. *[Target location(s), neighbourhood specification, function to apply.]*
- Q21.** Which raster overlay rule suits summing several binary presence/absence layers? *[Contributory rule.]*
- Q22.** What causes “scan skew” in satellite imagery? *[Mirror’s actual vs. nominal angular velocity varying during a scan.]*
- Q23.** Why must GCPs be well distributed, not clustered? *[Clustering biases the transform toward local accuracy and poor accuracy elsewhere, especially image edges.]*
- Q24.** Absolute vs. partial barrier in a friction surface? *[Absolute blocks passage entirely (detour around); partial allows passage at increased cost (rings compress/slow).]*
- Q25.** Between Clip and Intersect, which keeps attributes from BOTH layers? *[Intersect.]*
- Q26.** What makes an operation “Global” rather than “Focal”? *[It depends on processing the entire grid, not a bounded neighbourhood around one cell.]*
- Q27.** Vector vs. raster: which is “faster,” which is “corrector”? *[Raster is faster; vector is corrector – Joseph Berry.]*
- Q28.** Rasterization vs. vectorization – which is easier? *[Rasterization ($V \rightarrow R$) is easier; vectorization ($R \rightarrow V$) is harder.]*
- Q29.** What are the two decisions every choropleth map requires? *[Which attribute to map; how to classify its values.]*
- Q30.** Which classification method is best for showing “top 20% of districts”? *[Quantile (equal count).]*
- Q31.** What is the primary source of noise (low SNR) in satellite remote sensing? *[Atmospheric conditions – clouds, scattering.]*
- Q32.** Co-polarized vs. cross-polarized SAR – what does each detect best? *[Co-pol (HH/VV): surface roughness/man-made structures. Cross-pol (HV/VH): vegetation volume scattering.]*
- Q33.** IDW vs. Kriging? *[IDW: simple inverse-distance weighting. Kriging: statistical/geostatistical interpolation.]*
- Q34.** What is planar enforcement in topology? *[Polygons in a coverage must not overlap or leave gaps – every point belongs to exactly one polygon.]*